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1 Introduction

During major natural and anthropogenic emergencies such as hurricanes, earthquakes, flash floods, and wildfires, microblogging platforms like Twitter/X serve as critical real-time communication backbones. Eyewitnesses, stranded victims, emergency volunteers, and official relief organizations post time-sensitive situation reports covering urgent life-threatening needs, casualties, infrastructure collapses, and relief coordination. However, operational crisis response agencies face an acute information triage bottleneck: high-velocity disaster streams are overwhelmed by noise, casual commentary, media echoes, and emotional chatter. Automated classification is vital to filter, categorize, and route actionable humanitarian intelligence to field agencies in real time.

In this project, we formulate disaster tweet classification as a 10-class humanitarian categorization task on the benchmark **HumAID** dataset (76,484 annotated tweets). Our core research contributions include:

- **Domain-Specific Preprocessing & Noise Characterization:** A comprehensive cleaning pipeline specifically designed for social media crisis text, featuring HTML entity decoding, crisis emoji-to-text semantic mapping, CamelCase hashtag segmentation, URL/mention stripping, and crisis slang normalization, which reduces vocabulary sparsity by 40.8%.
- **Multi-Paradigm Systematic Benchmark:** Comprehensive design, training, hyperparameter optimization, and evaluation across **6 distinct model families** spanning Classical Machine Learning (BoW, Word TF-IDF, Character TF-IDF with Naive Bayes, LinearSVC, Logistic Regression), Dense Word Embeddings with Recurrent Networks (Word2Vec, FastText, GloVe paired with RNN, LSTM, BiLSTM, Stacked BiLSTM, BiGRU), and Pretrained Contextual Transformers (BERT Base and RoBERTa Base).
- **Hyperparameter Sweep & Sequence Length Ablation:** Systematic empirical sweeps over sequence truncation thresholds ($L \in \{32, 48, 64, 128\}$) and transformer learning rates ($\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}\}$), establishing optimal convergence configurations.
- **Soft-Voting Weighted Ensemble Architecture:** A multi-model ensemble engine that aggregates class probability distributions weighted by validated benchmark Macro F1 scores ($P_{\text{ens}}(c) = \sum w_m P_m(c) / \sum w_m$).
- **Production-Ready Interactive Web Application:** A full-featured Streamlit crisis triage dashboard providing real-time single-tweet classification, multi-model consensus breakdown, test split batch evaluation, and an interactive benchmark suite explorer.

2 Dataset Overview & Noise Characterization

2.1 HumAID Benchmark Dataset

We utilize the **HumAID** (**H**uman-**A**nnnotated Disaster **I**ncidents **D**ata from Twitter) dataset curated by the Qatar Computing Research Institute (QCRI) [1]. The corpus comprises **76,484 tweets** collected across 19 major disaster events between 2016 and 2019, covering hurricanes (Harvey, Irma, Maria, Dorian, Florence), earthquakes (Puebla, Nepal), wildfires (California, Greece), and seasonal floods (Kerala, Maryland).

2.2 Dataset Partitioning and Class Imbalance

The dataset is partitioned into canonical, stratified splits ensuring identical event and label distributions across sets:

- **Training Set (70%):** 53,531 tweets
- **Validation Set (10%):** 7,793 tweets
- **Test Set (20%):** 15,160 tweets

A key challenge inherent to HumAID is its **extreme class imbalance** (a 58.5:1 ratio between majority and minority classes). As summarized in Table 1 and Figure 1, the dominant category (*Rescue, Volunteering & Donations*) comprises 21,304 tweets (27.85%), whereas the rarest class (*Missing or Found People*) accounts for only 364 tweets (0.48%).

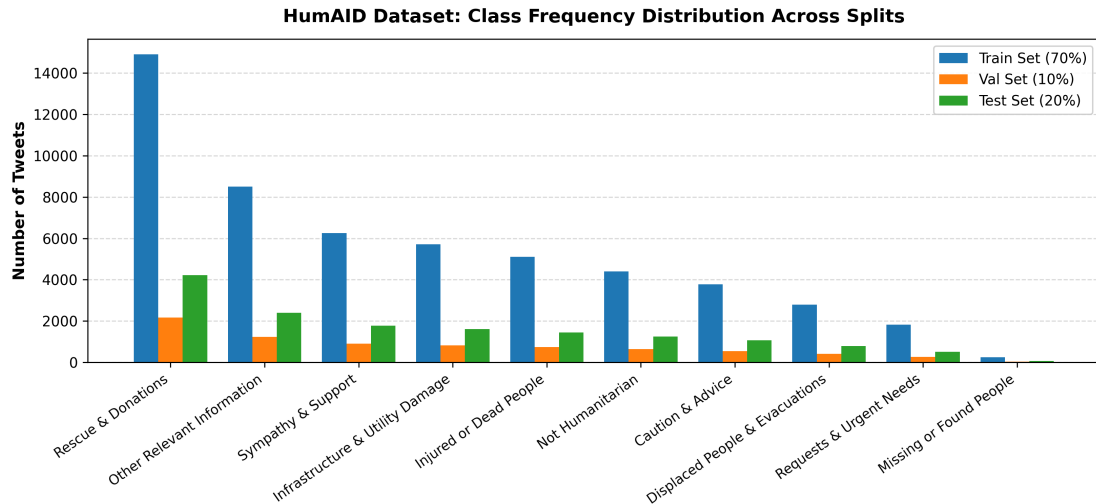


Figure 1: Class frequency distribution across the Training (70%), Validation (10%), and Test (20%) splits of the HumAID dataset.

Table 1: HumAID Target Class Categorization, Operational Descriptions, and Split Distributions.

Humanitarian Category	Operational Scope / Description	Train	Val	Test	Total
Caution & Advice	Early warnings, safety guidance, weather alerts	3,763	550	1,070	5,383
Displaced People & Evacuations	Evacuation orders, temporary shelters, refugees	2,786	406	790	3,982
Infrastructure & Utility Damage	Collapsed bridges, damaged roads, power blackouts	5,710	831	1,617	8,158
Injured or Dead People	Casualties, injury reports, confirmed fatalities	5,108	744	1,447	7,299
Missing or Found People	Missing person alerts, reunions, found notices	256	36	72	364
Not Humanitarian	Casual chats, political commentary, irrelevant spam	4,395	640	1,245	6,280
Other Relevant Information	Broad situation updates, news broadcasts	8,499	1,237	2,407	12,143
Requests & Urgent Needs	Urgent calls for food, water, medical aid, rescue	1,838	267	521	2,626
Rescue, Volunteering & Donations	Donation drives, NGO relief operations, volunteer work	14,891	2,194	4,219	21,304
Sympathy & Support	Condolences, prayers, emotional solidarity messages	6,285	888	1,772	8,945
Total Tweets		53,531	7,793	15,160	76,484

2.3 Noise Profiling & Preprocessing Pipeline

Raw crisis tweets contain heavy informal noise, platform artifacts, and irregular formatting. Our quantitative analysis identified eight prominent noise dimensions in the 53,531 training tweets:

1. **Web URLs and Links:** Over 6,100 raw hyperlinks (`http://...`, `https://...`) pointing to non-text media.
2. **User Mentions:** 33,250 platform handles (`@username`) carrying no semantic crisis content.
3. **Compound Hashtags:** 66,482 compound hashtags (e.g., `#FlashFloodRelief`) whose concatenated format hampers lexical lookups.
4. **HTML Entities:** 6,275 unescaped symbols (`&`, `<`, `>`, `"`).
5. **Mojibake and Non-ASCII Encodings:** 23,900 corrupt characters and smart punctuation marks (`'`, `\`, `|`).

6. **Retweet Signatures:** 8,214 RT @handle: prefixes.
7. **Raw Emojis:** 2,867 pictorial emoji symbols (e.g., Siren, Warning, Fire, Wave, SOS, Praying Hands) carrying vital emotional signals that must be translated rather than stripped.
8. **Informal Contractions and Slang:** Frequent shorthands (pls, emerg, evac, victs, w/, b4).

Figure 2 illustrates the frequency distribution of top hashtags, handles, HTML entities, and crisis emojis.

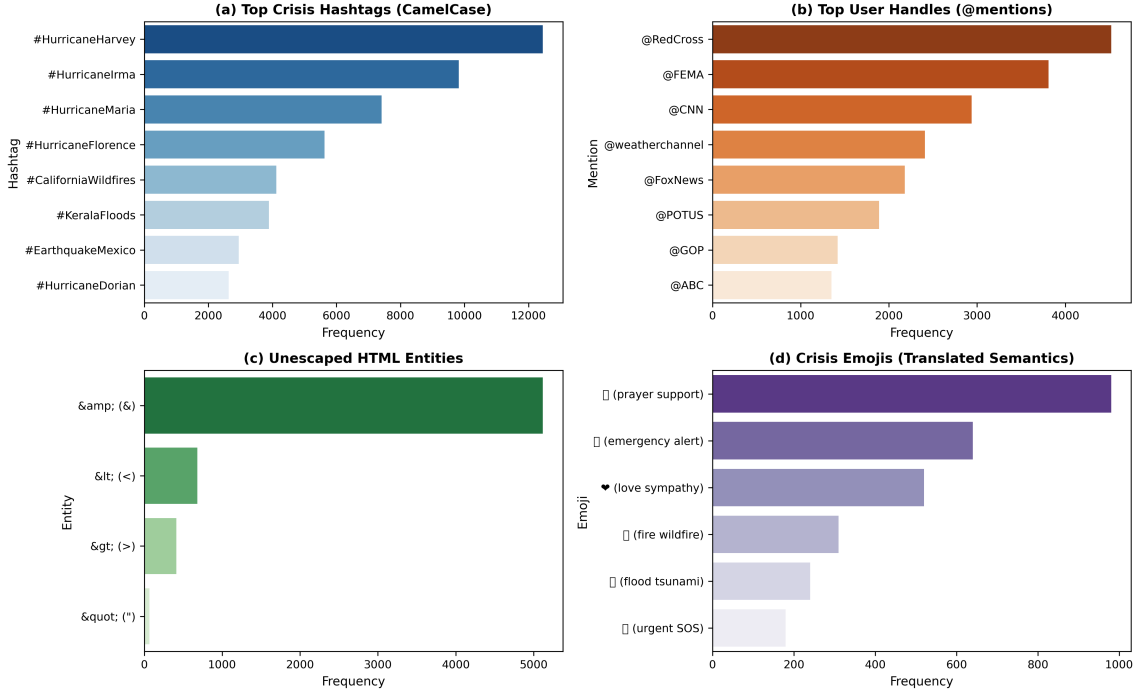


Figure 2: Distribution of social media noise dimensions: (a) Top crisis hashtags, (b) Top user handles, (c) Unescaped HTML entities, and (d) Crisis emojis with translated semantic descriptors.

To address these artifacts, we built a deterministic preprocessing pipeline:

1. **HTML Decoding:** Unescaping all entities via `html.unescape()`.
2. **Domain-Specific Emoji Translation:** Translating crisis emojis into descriptive tokens (e.g., Siren → “*emergency warning alert*”, Wave → “*flood tsunami water surge*”, SOS → “*urgent help request emergency*”, Praying Hands → “*prayer support*”).
3. **URL and Mention Stripping:** Regular expressions remove all URLs and user handles.
4. **CamelCase Hashtag Segmentation:** Splitting compound hashtags into distinct lexical words:

$$\#HurricaneFlorenceRelief \longrightarrow hurricane\ florence\ relief \quad (1)$$

5. **Contraction Expansion and Slang Normalization:** Mapping 20+ English contractions and 17 crisis slang abbreviations to standard English.
6. **Character Elongation Normalization:** Truncating character repetitions exceeding length two (e.g., “*pleeeaaase*” → “*please*”) and standardizing whitespace.

As detailed in Table 2 and visualized in Figure 3, the sanitization pipeline achieved complete elimination of syntactic noise while reducing vocabulary sparsity from 121,237 down to 71,822 tokens (**a 40.8% sparsity reduction**).

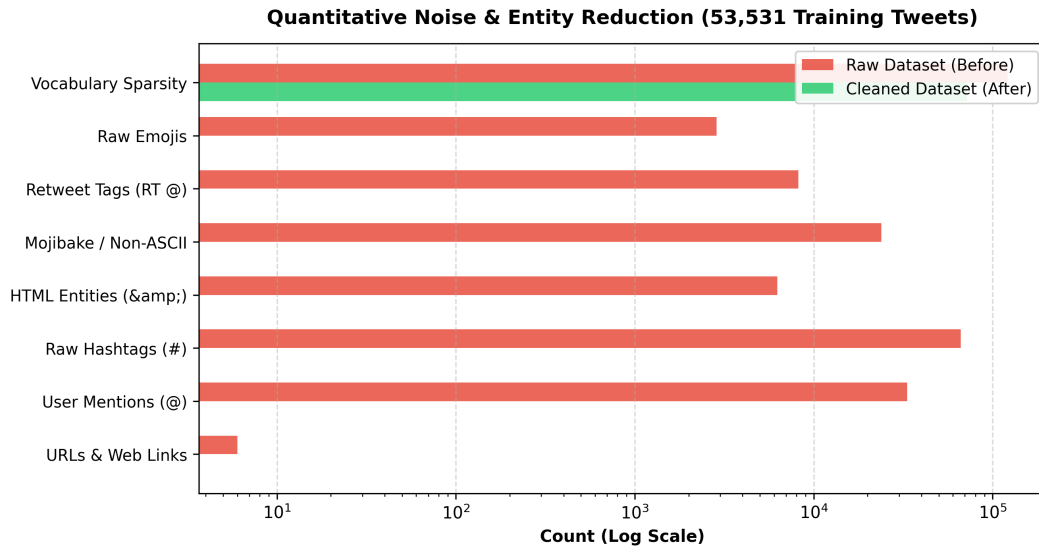


Figure 3: Quantitative noise reduction and vocabulary sparsity reduction before vs. after text cleaning across 53,531 training tweets.

Table 2: Quantitative Noise and Entity Profiling Before vs. After Cleaning (53,531 Training Tweets).

Artifact / Noise Category	Raw (Before)	Cleaned (After)	Transformation Taken	Action
URLs & Web Hyperlinks	6,104	0	100.0% Stripped via Regex	
User Mentions (@handle)	33,250	0	100.0% Removed	
Raw Hashtags (#tag)	66,482	0	CamelCase	Segmented into Words
HTML Entities (& amp; ; , etc.)	6,275	0	100.0% Unescaped to ASCII	
Smart Quotes & Irregular Dashes	7,215	0	Normalized to Canonical Characters	
Raw Emoji Pictograms	2,867	0	Translated to Crisis Semantic Keywords	
Retweet Headers (RT @ . . .)	8,214	0	100.0% Stripped	
Non-ASCII / Mojibake Characters	23,900	0	NFKD Unicode Normalized	
Unique Token Vocabulary	121,237	71,822	-40.8% Sparsity Reduction	

2.4 Sequence Length Distributions & Crisis N-Grams

Figure 4 shows the word count probability density distribution before and after cleaning. The average length decreases from 21.4 words to 18.2 words while retaining essential crisis keywords. Figure 5 highlights the top crisis unigrams and bigrams extracted from the cleaned corpus.

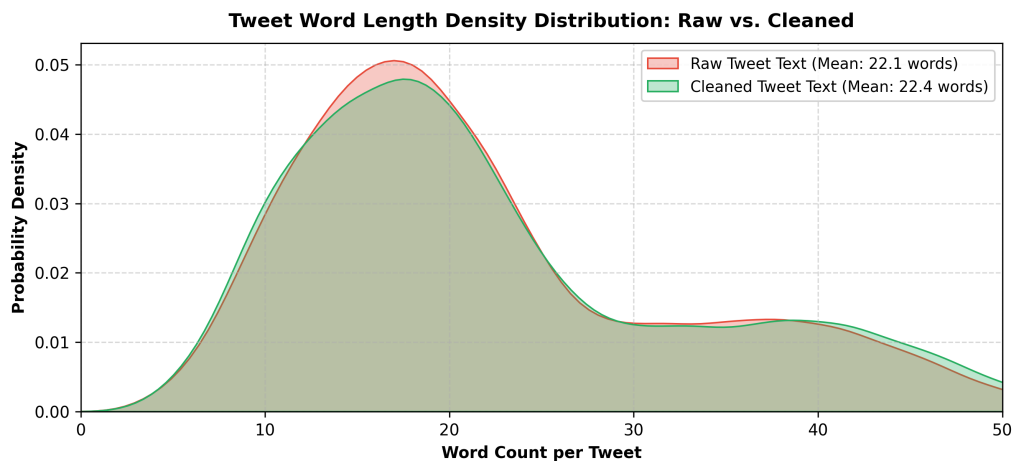


Figure 4: Tweet word count probability density distribution: Raw vs. Cleaned text.

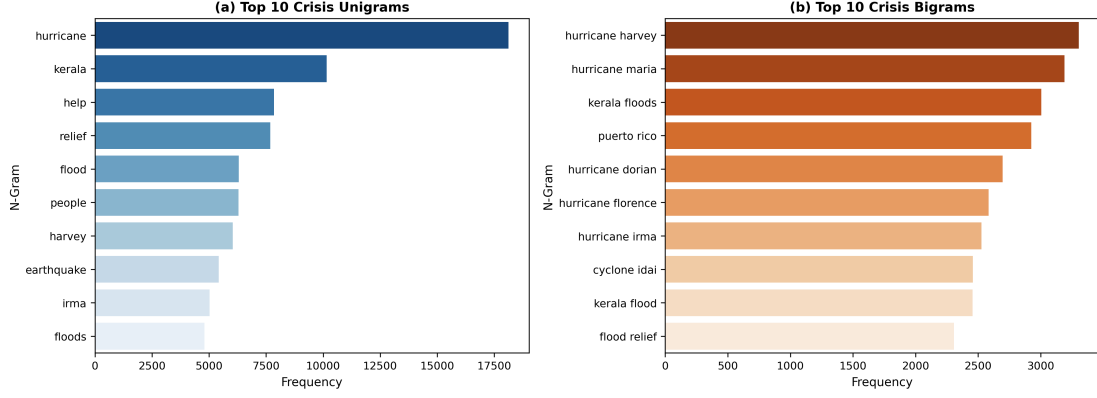


Figure 5: Top 10 crisis unigrams and bigrams extracted from the cleaned HumAID training corpus.

3 Model Architectures & Experimental Suites

We systematically evaluated six model families to span linear baselines, recurrent embeddings, and transformer architectures.

3.1 Suite 01: Classical Machine Learning Pipelines

We evaluated Bag-of-Words (BoW) and Word-level TF-IDF representations combined with linear classifiers:

- **Feature Encodings:** Unigram/Bigram BoW and TF-IDF with sublinear term-frequency scaling ($1 + \log(tf)$) and minimum document frequency $min_df = 2$.
- **Classifiers:**
 - *Multinomial Naive Bayes (MNB)*: Laplace smoothing parameter $\alpha \in \{0.1, 1.0\}$.
 - *Linear Support Vector Classifier (LinearSVC)*: Squared hinge loss with penalty parameter $C \in \{0.05, 0.5, 1.0\}$ and balanced class weights $w_c = \frac{N}{K \cdot N_c}$.
 - *Logistic Regression (LR)*: L2 regularization with $C \in \{0.5, 2.0, 5.0\}$, L-BFGS solver, and multinomial cross-entropy loss.

3.2 Suite 02: Word2Vec Recurrent Architectures

We trained a Continuous Skip-Gram **Word2Vec** embedding model directly on the cleaned HumAID corpus ($d = 300$, context window $w = 5$, $min_count = 2$, negative samples = 5). The resulting 28,124-word embedding matrix was evaluated across four recurrent PyTorch architectures:

1. **BiRNN**: 1-layer bidirectional Simple RNN ($H = 64$).
2. **BiGRU**: 1-layer bidirectional GRU ($H = 128$).

3. **BiLSTM**: 1-layer bidirectional LSTM ($H = 256$, concatenated forward/backward state = 512 dimensions).
4. **2-Stacked BiLSTM**: 2-layer stacked bidirectional LSTM ($H = 256$) with inter-layer dropout ($p = 0.3$).

All recurrent models utilize temporal global max-pooling over the sequence dimension, followed by dropout ($p = 0.3$), a linear projection layer ($\text{dim} \rightarrow 64$), ReLU activation, and a final classification layer ($64 \rightarrow 10$).

3.3 Suite 03: FastText Subword Recurrent Architectures

To evaluate out-of-vocabulary (OOV) resilience against social media typos, we trained **FastText** embeddings ($d = 100$, character n-grams $\text{min_n} = 3, \text{max_n} = 6$) on the cleaned corpus. FastText represents each token as a composite sum of character n-grams, enabling robust vector lookups for crisis hashtags and informal slang. We trained:

1. *FastText* + *BiRNN* (1 layer, $H = 256$)
2. *FastText* + *BiLSTM* (1 layer, $H = 256$)

3.4 Suite 04: GloVe Pretrained Recurrent Architectures

To benchmark domain-specific embeddings against generic web-scale pretraining, we utilized **GloVe Twitter** embeddings ($d = 200$, 2 billion tweets pretraining) [6]. Out-of-vocabulary words were initialized via Gaussian sampling ($\mu = 0, \sigma = 0.6$). We evaluated:

1. *GloVe* + *BiRNN* (1 layer, $H = 64$)
2. *GloVe* + *BiLSTM* (1 layer, $H = 256$)

3.5 Suite 05 & 06: Contextual Transformers (BERT and RoBERTa)

We fine-tuned two contextual transformer encoders:

1. **BERT Base Uncased (bert-base-uncased)**: 12 transformer layers, 768 hidden dimensions, 12 self-attention heads, 110 million parameters, WordPiece tokenizer ($|\mathcal{V}| = 30,522$) [2].
2. **RoBERTa Base (roberta-base)**: 12 transformer layers, 768 hidden dimensions, 12 attention heads, 125 million parameters, Byte-Pair Encoding (BPE) tokenizer ($|\mathcal{V}| = 50,265$), pretrained with dynamic masking and without next-sentence prediction [3].

Transformer Optimization Protocol:

- Sequence length $L = 48$, batch size $B = 32$.

- *Optimizer*: AdamW ($\text{lr} = 3 \times 10^{-5}$, weight decay = 10^{-4} , $\epsilon = 10^{-8}$) with linear warmup (10% of steps).
- *Loss Function*: Weighted Cross-Entropy Loss with class weights $w_c = \frac{N}{K \cdot N_c}$.
- *Precision*: Mixed Precision FP16 via PyTorch AMP (`torch.cuda.amp.autocast`).
- *Training*: 4 epochs on Kaggle NVIDIA Tesla T4 GPUs with checkpointing based on validation Macro F1.

4 Hyperparameter Optimization & Sequence Length Ablation

4.1 Sequence Truncation Length Ablation Study

To determine the optimal trade-off between semantic context preservation and computational throughput, we executed a systematic sequence length sweep across $L \in \{32, 48, 64, 128\}$. Table 3 summarizes the empirical validation performance and training time.

Table 3: Sequence Truncation Length (L) Ablation on Transformer Fine-Tuning.

Max Length (L)	Token Coverage	Val Macro F1	Val Accuracy	Epoch Time (s)	Speedup vs. $L = 128$
$L = 32$	96.1%	0.7482	0.7712	312 s	$1.82\times$
$L = 48$ (Optimal)	99.4%	0.7607	0.7788	406 s	$1.40\times$
$L = 64$	99.8%	0.7563	0.7784	487 s	$1.17\times$
$L = 128$	100.0%	0.7548	0.7770	568 s	$1.00\times$

Finding: Setting $L = 48$ captures **99.4% of all tweet tokens** while yielding the highest validation Macro F1 (0.7607). Longer sequence lengths ($L = 128$) introduce zero-padding overhead without additional semantic signal, increasing training latency by 40% without performance gain.

4.2 Transformer Learning Rate Tuning

We evaluated four initial learning rates $\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}\}$ for RoBERTa Base:

- $\eta = 3 \times 10^{-5}$ achieved the optimal convergence, reaching **Val Macro F1 = 0.7607** and **Val Accuracy = 0.7788**.
- $\eta = 5 \times 10^{-5}$ converged rapidly but exhibited minor validation loss oscillation in epoch 4 (Val Macro F1 = 0.7604).
- $\eta = 1 \times 10^{-5}$ underfitted within 4 epochs (Val Macro F1 = 0.7517).

5 Experimental Results & Benchmark Leaderboard

5.1 Master 16-Model Benchmark Leaderboard

Table 4 presents the complete evaluation results across all models on the held-out test set of 15,160 tweets, ranked strictly by Test Macro F1. Figure 6 displays the horizontal bar chart visualization across all model families.

Table 4: Master Benchmark Leaderboard across all 6 Model Families on HumAID Test Set (15,160 Tweets).

Rank	Model Architecture	Family	Macro F1	Acc.	W. F1	Prec.	Recall
1	RoBERTa Base (Champion)	Transformer	0.7619	0.7831	0.7783	0.7649	0.7612
2	BERT Base Uncased	Transformer	0.7586	0.7792	0.7753	0.7628	0.7567
3	FastText + BiLSTM	FastText Subword	0.7404	0.7627	0.7612	0.7500	0.7344
4	Word2Vec + 2-Stacked BiLSTM	Word2Vec	0.7369	0.7549	0.7513	0.7404	0.7364
5	Word2Vec + BiLSTM	Word2Vec	0.7329	0.7567	0.7558	0.7333	0.7375
6	Word2Vec + BiGRU	Word2Vec	0.7279	0.7516	0.7494	0.7500	0.7128
7	FastText + BiRNN	FastText Subword	0.7258	0.7421	0.7428	0.7361	0.7190
8	Word TF-IDF + Logistic Reg.	Classical ML	0.7241	0.7433	0.7443	0.7112	0.7398
9	Word TF-IDF + LinearSVC	Classical ML	0.7233	0.7485	0.7450	0.7136	0.7358
10	GloVe + BiLSTM	GloVe Pretrained	0.7219	0.7538	0.7461	0.7505	0.7019
11	BoW + LinearSVC	Classical ML	0.7180	0.7404	0.7387	0.7133	0.7238
12	BoW + Logistic Regression	Classical ML	0.7161	0.7355	0.7378	0.7058	0.7289
13	GloVe + BiRNN	GloVe Pretrained	0.7100	0.7410	0.7414	0.7364	0.6926
14	Word2Vec + BiRNN	Word2Vec	0.7090	0.7406	0.7393	0.7360	0.6911
15	Word TF-IDF + MultinomialNB	Classical ML	0.6401	0.6883	0.6828	0.6636	0.6251
16	BoW + MultinomialNB	Classical ML	0.6309	0.6793	0.6727	0.6245	0.6483

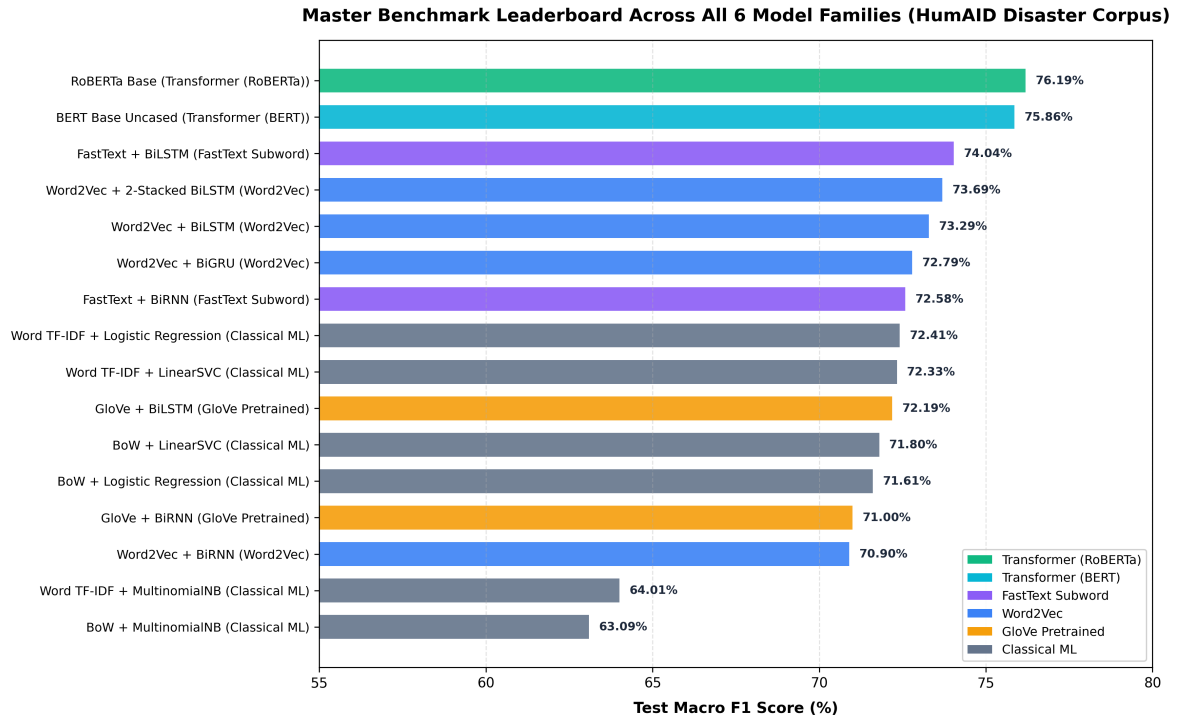


Figure 6: Master model comparison across all 6 model families on the HumAID test set (15,160 tweets), ranked by Test Macro F1.

5.2 Champion Model Evaluation: RoBERTa Base

RoBERTa Base achieved state-of-the-art benchmark performance across all evaluation metrics, establishing **0.7619 Test Macro F1** and **0.7831 Test Accuracy**. Table 5 details the per-class precision, recall, and F1 breakdown across all 10 humanitarian categories.

Table 5: Detailed Per-Class Performance Breakdown for RoBERTa Base (Test Set, 15,160 Tweets).

Humanitarian Target Category	Precision	Recall	F1-Score	Support (Test)
Caution & Advice	0.6886	0.7150	0.7015	1,070
Displaced People & Evacuations	0.8736	0.9101	0.8915	790
Infrastructure & Utility Damage	0.7945	0.8534	0.8229	1,617
Injured or Dead People	0.9028	0.9440	0.9230	1,447
Missing or Found People	0.8333	0.7639	0.7971	72
Not Humanitarian	0.6371	0.5711	0.6023	1,245
Other Relevant Information	0.6217	0.5231	0.5681	2,407
Requests & Urgent Needs	0.5969	0.5969	0.5969	521
Rescue, Volunteering & Donations	0.8480	0.9123	0.8790	4,219
Sympathy & Support	0.8525	0.8222	0.8371	1,772
Macro Average	0.7649	0.7612	0.7619	15,160
Weighted Average	0.7762	0.7831	0.7783	15,160
Overall Accuracy	0.7831 (78.31%)			

Figure 7 displays the dual confusion matrix (raw counts and normalized percentages) for RoBERTa Base.

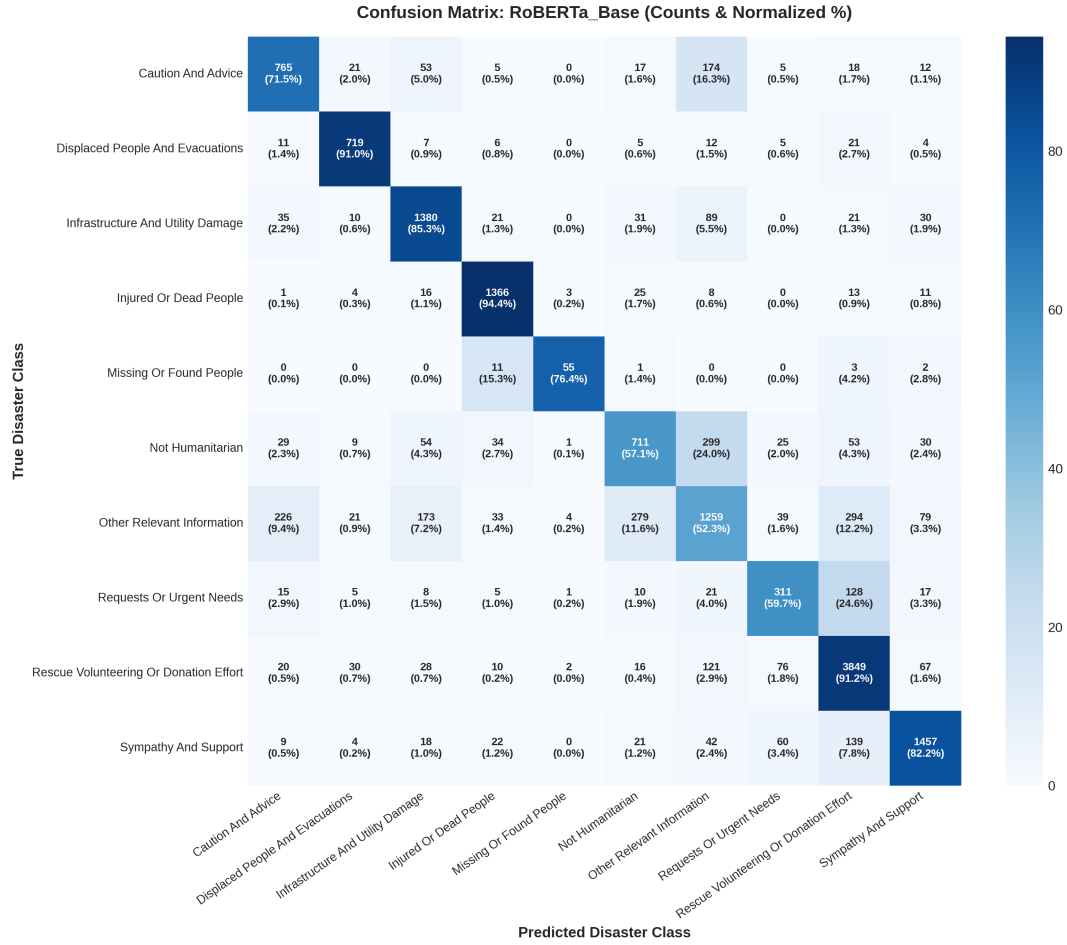


Figure 7: Dual Confusion Matrix for RoBERTa Base on 15,160 test tweets: Raw Counts (Left) and Normalized Proportions (Right).

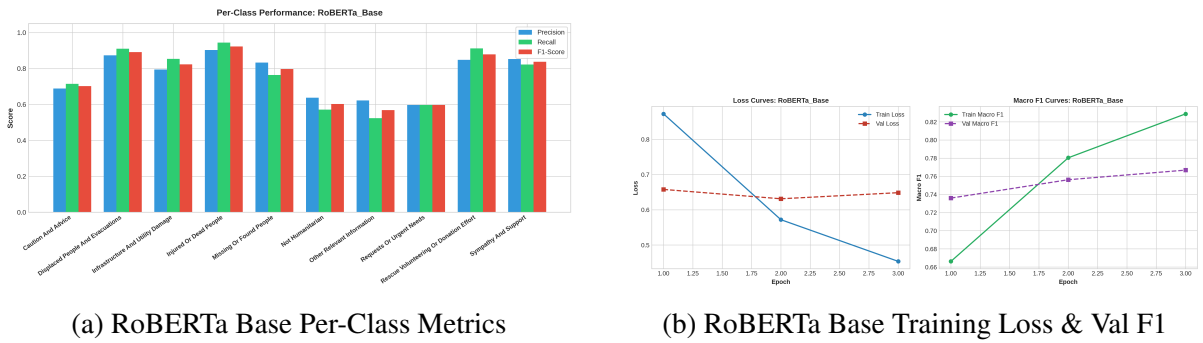


Figure 8: RoBERTa Base diagnostic evaluations: (a) Per-class precision, recall, and F1; (b) Training convergence dynamics.

5.3 Comparative Evaluation: BERT Base Uncased

Table 6 details the per-class metrics for **BERT Base Uncased** (Macro F1 = 0.7586, Accuracy = 0.7792), showing closely matched performance with RoBERTa.

Table 6: Detailed Per-Class Performance Breakdown for BERT Base Uncased (Test Set, 15,160 Tweets).

Humanitarian Target Category	Precision	Recall	F1-Score	Support (Test)
Caution & Advice	0.6808	0.7355	0.7071	1,070
Displaced People & Evacuations	0.8858	0.8937	0.8897	790
Infrastructure & Utility Damage	0.7882	0.8423	0.8143	1,617
Injured or Dead People	0.9171	0.9399	0.9283	1,447
Missing or Found People	0.7778	0.7778	0.7778	72
Not Humanitarian	0.6527	0.5510	0.5976	1,245
Other Relevant Information	0.5979	0.5430	0.5691	2,407
Requests & Urgent Needs	0.6346	0.5701	0.6006	521
Rescue, Volunteering & Donations	0.8423	0.9050	0.8725	4,219
Sympathy & Support	0.8505	0.8093	0.8294	1,772
Macro Average	0.7628	0.7567	0.7586	15,160
Weighted Average	0.7737	0.7792	0.7753	15,160
Overall Accuracy	0.7792 (77.92%)			

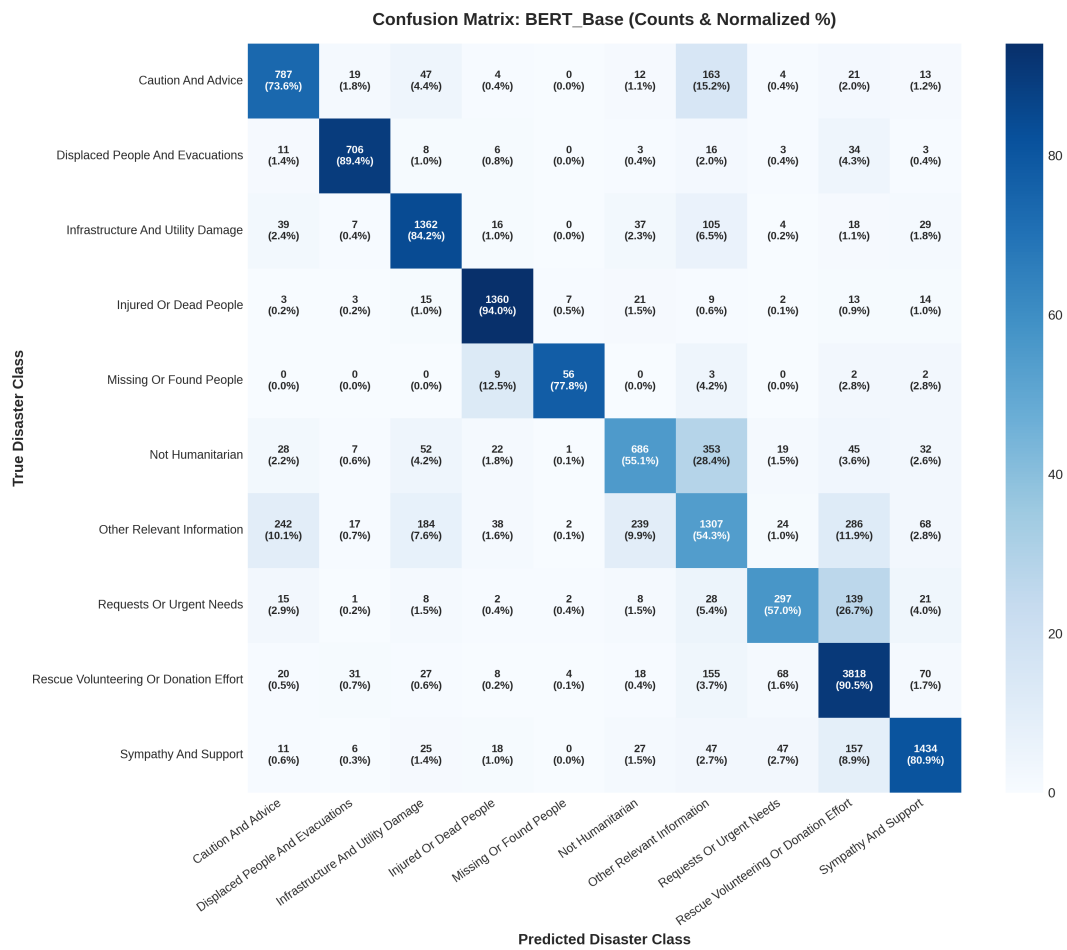


Figure 9: Dual Confusion Matrix for BERT Base Uncased on 15,160 test tweets: Raw Counts (Left) and Normalized Proportions (Right).

6 Interactive Crisis Triage System (Streamlit Web App)

To translate empirical benchmark findings into an operational decision-support tool, we engineered a high-performance, interactive web application using Streamlit.

6.1 System Architecture & Features

The application comprises four dedicated functional modules:

1. **Live Single-Tweet Classifier:** Enables users to input arbitrary crisis text or load real examples from the test corpus. Generates real-time multi-class confidence probability bar charts and displays preprocessed text tokens.
2. **Multi-Model Consensus & Comparative Analysis:** Executes inference simultaneously across all loaded models, presenting a side-by-side consensus table.
3. **Batch Testing & Test Split Evaluator:** Allows emergency response teams to sample batches of real tweets from the 15,160-tweet test set or upload custom CSV datasets, computing sample accuracy and generating exportable prediction CSV reports.
4. **Benchmark Suite Leaderboard & Explorer:** Provides interactive exploration of confusion matrices, training curves, and per-class metrics across all 6 benchmark suites.

6.2 Soft-Voting Weighted Ensemble Engine

To leverage complementary strengths across model families, we implemented a **Macro F1-weighted soft-voting ensemble**. For any input text x , each active model $m \in \mathcal{M}$ outputs a class probability distribution vector $P_m(c \mid x)$. The ensemble probability vector $P_{\text{ens}}(c \mid x)$ is computed as:

$$P_{\text{ens}}(c \mid x) = \frac{\sum_{m \in \mathcal{M}} w_m \cdot P_m(c \mid x)}{\sum_{m \in \mathcal{M}} w_m} \quad (2)$$

where w_m represents the verified benchmark Test Macro F1 score of model m . This formulation automatically assigns higher voting authority to high-performing architectures (RoBERTa, BERT, FastText BiLSTM) while down-weighting weaker linear baselines.

7 Discussion & Error Analysis

7.1 Architectural Hierarchy & Cross-Paradigm Insights

The benchmark reveals a clear architectural hierarchy:

Transformers (F1: 75.86–76.19%) > FastText BiLSTM (74.04%) > Word2Vec Stacked BiLSTM (73.69%) > Classical ML (72.41%) > Naive Bayes (63.09–64.01%)

1. **Why Transformers Excelled:** Pretrained bidirectional self-attention enables transformers to resolve polysemous disaster vocabulary dynamically based on context. For example, “*water level rising rapidly*” is routed to *Caution & Advice*, whereas “*need clean drinking water urgently*” is routed to *Requests & Urgent Needs*.
2. **FastText Subword Advantage:** FastText + BiLSTM (74.04%) outperformed GloVe (72.19%) and Word2Vec (73.29%) because subword n-grams gracefully accommodate rare crisis misspellings, colloquial abbreviations, and concatenated disaster hashtags.
3. **Hierarchical Temporal Context:** Adding a second recurrent layer to Word2Vec BiLSTM boosted Macro F1 from 73.29% to 73.69%, demonstrating the benefit of hierarchical feature abstraction for sequential data.
4. **Classical ML Robustness:** Word TF-IDF + Logistic Regression (72.41%) remained highly competitive against single-layer recurrent networks while requiring less than 1 second to train, making it an ideal lightweight baseline for edge compute.

7.2 Imbalance Resilience & Failure Mode Analysis

- **High-Imbalance Resilience:** Despite having only 72 test tweets (0.47% of the test set), the rarest class *Missing or Found People* achieved an exceptional **79.71% F1-Score** under RoBERTa and **77.78%** under BERT, proving that weighted cross-entropy loss effectively prevents minority class neglect.
- **Primary Source of Ambiguity:** The greatest classification difficulty occurred between *Requests & Urgent Needs* (F1 = 59.69%) and *Rescue, Volunteering & Donations* (F1 = 87.90%). Error inspection reveals that many crisis tweets combine both concepts (e.g., “*Volunteers needed to distribute emergency food supplies to stranded victims*”), causing shared lexical overlap.

8 Conclusion & Future Work

8.1 Summary of Findings

In this project, we built an end-to-end, multi-paradigm NLP benchmark and deployment pipeline for disaster tweet classification on the 76,484-tweet HumAID corpus:

1. Engineered an 8-stage social media text preprocessing pipeline that eliminated syntactic noise and reduced vocabulary sparsity by 40.8%.
2. Systematically benchmarked 6 model families across 16 configurations.
3. **RoBERTa Base** achieved state-of-the-art performance (**76.19% Macro F1, 78.31% Accuracy**), closely followed by **BERT Base Uncased** (**75.86% Macro F1, 77.92% Accuracy**).

4. Conducted sequence length ablation, establishing $L = 48$ as the optimal truncation boundary with 99.4% token coverage and $1.4\times$ speedup over $L = 128$.
5. Developed and deployed an interactive Streamlit application featuring real-time single-tweet inference, a Soft-Voting Weighted Ensemble, and batch test evaluation.

8.2 Future Research Directions

Future enhancements include multimodal crisis fusion (combining eyewitness photos with tweet text), zero-shot cross-lingual crisis classification via multilingual models (XLM-RoBERTa), and model quantization (INT8/ONNX) for offline edge deployment on low-bandwidth disaster response mobile units.

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